Lecture 21-22 Scribe

L21_L22.pdf
PDF

lecture1.pdf
PDF

EXAMPLE LECTURE SCRIBE (STRUCTURE REFERENCE):
lecture1.pdf

INSTRUCTION:
Using the exact same structure and formatting as the example scribe above, generate a comprehensive lecture scribe for L21-22 based on the provided lecture materials.

PURPOSE:
This scribe will serve as a student's primary exam reference document. A student studying only this document should be able to:
- Recall all definitions, notation, and conventions introduced
- Reconstruct every proof and derivation step-by-step
- Follow all example solutions with clear intermediate calculations
- Understand how concepts build upon each other
- Locate any result or explanation quickly

CONTENT RULES:
- Include ONLY what appears in the provided lecture materials
- Include ALL definitions, theorems, proofs, derivations, and worked examples exactly as presented
- Include EVERY example from the lecture with complete step-by-step solutions
- Do NOT add external explanations, alternative proofs, or supplementary examples
- Do NOT add commentary, study tips, or explanatory notes beyond what's in the lecture
- Do NOT simplify or rephrase proofs—preserve the original level of detail
- Do NOT omit any examples, even if they seem repetitive

STRUCTURE REQUIREMENTS:

Use clear section headings for each major topic. For each component:

Definitions and Notation:
- Precise mathematical statements with symbol definitions
- Domain, range, and any assumptions/conditions

Theorems and Results:
- Clear statement separating hypothesis and conclusion
- Step-by-step proof with each step justified
- References to definitions used

Worked Examples:
- EVERY example from the lecture must appear
- Problem statement followed by step-by-step solution
- ALL intermediate calculations shown—no skipped steps
- Final answer clearly indicated
- Label examples sequentially (Example 1, Example 2, etc.)

Derivations:
- Show each algebraic or logical step
- Include brief justifications for non-obvious steps

Logical Flow:
- "Recall from previous section..." statements where appropriate
- "This leads to..." transitions showing dependencies
- Prerequisite concepts noted before new material

FORMATTING:
- Match the example's heading hierarchy and indentation
- Use consistent notation conventions
- Preserve any visual structuring (boxes, spacing, etc.)

MATERIALS TO PROCESS:
L21-22.pdf

Writing

CSE400 — Fundamentals of Probability in Computing

Lecture 21–22: Inequalities and Tail Bounds

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1. Introduction

These notes summarize the technical content of Lectures 21–22. The focus is on **tail bounds** and fundamental inequalities used to bound probabilities of extreme events.

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2. Motivation: Tail Bounds

Definition (Tail Probability)

For a random variable (X), the **tail probability** refers to:
[
\Pr(X \geq t) \quad \text{or} \quad \Pr(|X - \mathbb{E}[X]| \geq t)
]

Purpose

- Quantify **rare events**
- Bound **worst-case deviations**
- Used in **system design and reliability analysis**

Transition

This leads to inequalities that bound these probabilities using limited information (e.g., expectation, variance).

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3. Markov's Inequality

Definition

Let (X \geq 0) be a non-negative random variable. Then for any (a > 0):
[

\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}
]

Proof

Start with expectation:
[
\mathbb{E}[X] = \sum_x x \cdot \Pr(X = x)
]

Split the sum:
[
\mathbb{E}[X] \geq \sum_{x \geq a} x \cdot \Pr(X = x)
]

Since (x \geq a):
[
\mathbb{E}[X] \geq \sum_{x \geq a} a \cdot \Pr(X = x)
]

Factor out (a):
[
\mathbb{E}[X] \geq a \cdot \Pr(X \geq a)
]

Divide both sides:
[
\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}
]

Example 1

Problem:
Let (\mathbb{E}[X] = 10). Find an upper bound for (\Pr(X \geq 20)).

Solution:
[
\Pr(X \geq 20) \leq \frac{10}{20} = 0.5
]

Final Answer: (\boxed{0.5})

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4. Chebyshev's Inequality

Definition

For any random variable (X) with mean (\mu) and variance (\sigma^2):
[
\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}
]

Proof

Apply Markov's inequality to:
[
Y = (X - \mu)^2
]

Then:
[
\Pr((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2}
]

Since:
[
\mathbb{E}[(X - \mu)^2] = \sigma^2
]

We get:
[
\Pr(|X - \mu| \geq k\sigma) \leq \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}
]

Example 2

Problem:
Let (\mu = 50), (\sigma = 5). Find:

[
\Pr(|X - 50| \geq 10)
]

Solution:
[
k = \frac{10}{5} = 2
]

[
\Pr(|X - 50| \geq 10) \leq \frac{1}{2^2} = \frac{1}{4}
]

Final Answer: (\boxed{0.25})

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5. Chernoff Bound and Poisson Tail

Definition (Chernoff Bound)

For independent random variables, Chernoff bounds give exponentially decreasing bounds on tail probabilities.

General form:
[
\Pr(X \geq (1+\delta)\mu) \leq e^{-\text{function of } \delta \cdot \mu}
]

Key Idea

- Uses **moment generating functions**
- Provides **tighter bounds than Chebyshev**
- Applicable for **sum of independent variables**

Example 3

Problem:
Bound probability that sum deviates significantly from expectation.

Solution:
Apply Chernoff inequality:
[
\Pr(X \geq (1+\delta)\mu) \leq e^{-c \cdot \mu}
]

Final Answer: Exponential decay bound in terms of (\mu)

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6. Jensen's Inequality

Definition

Let (\phi) be a convex function and (X) a random variable. Then:
[
\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]
]

Convex Function

A function (\phi) is convex if:
[
\phi(\lambda x + (1-\lambda)y) \leq \lambda \phi(x) + (1-\lambda)\phi(y)
]

for (0 \leq \lambda \leq 1)

Proof (Sketch)

Using convexity:
[
\phi\left(\sum p_i x_i\right) \leq \sum p_i \phi(x_i)
]

Recognizing expectation:
[
\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]
]

Example 4

Problem:
Show:
[
(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]
]

Solution:
Let (\phi(x) = x^2), which is convex.

Apply Jensen:
[
(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]
]

Final Answer: Proven using Jensen's inequality

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7. Case Study: Predicting Worst-Case Demand

Problem

Estimate worst-case server demand using probabilistic bounds.

Approach

- Use **Markov** for coarse bounds
- Use **Chebyshev** when variance known
- Use **Chernoff** for tight exponential bounds

Interpretation

- Tail bounds help determine **capacity planning**
- Provide **guarantees on rare events**

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8. Summary

- Tail bounds quantify extreme deviations
- Markov uses expectation
- Chebyshev uses variance

- Chernoff provides exponential decay
- Jensen connects convexity with expectation

End of Lecture 21–22